

Problem Set 2

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Problem 1. Asymptotic Relationships

(a)

$f(n)$	$g(n)$	Relationship?
6^n	n^{1200}	$f \in \omega(g)$
$(\log n)^6$	$\sqrt[1200]{n}$	$f \in o(g)$
$\log_6(n^{1200})$	$(\log_{1200} n)^6$	$f \in o(g)$
$\log(n!)$	$n \log n$	$f \in \Theta(g)$
$2^{\lceil \log_2 n \rceil}$	$3^{\lceil \log_3 n \rceil}$	$f \in o(g)$
$2^{\lceil \log_2 n \rceil} - n + 1$	$3^{\lceil \log_3 n \rceil} - n + 1$	$f \in o(g)$
n	$n^{\cos \log n}$	None.

(b)

$f(n)$	$g(n)$	Relationship?	Justification
$2^{(\log n)^2}$	$2^{(\lceil \log n \rceil)^2}$	None.	$\lim_{x \rightarrow \infty} \frac{f(n)}{g(n)}$ does not exist
4^n	$(2n)!/(n!)^2$	$f \in \omega(g)$	$\lim_{x \rightarrow \infty} \frac{f(n)}{g(n)} = \infty$

Note: For equation $\frac{2^{(\log n)^2}}{2^{(\lceil \log n \rceil)^2}}$ limit does not exist because at infinity it's value oscillates between 0 and 1.

Problem 2. OEIS

(a)

$$x_3 = 5, x_4 = 12, x_5 = 29, x_6 = 70$$

(b)(c) Found formula:

$$a(n) = \frac{(1 + \sqrt{2})^n - (1 - \sqrt{2})^n}{2\sqrt{2}}$$

(d)

Proof by strong induction. Induction hypothesis which says, sequence that begins with $x_0 = 0, x_1 = 1, x_2 = 2$ and continues with $x_n = 5x_{n-2} + 2x_{n-3}$ for $n \geq 3$, nth term can also be calculated via $P(n) = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}$ for $x \geq 0$

Base cases: $P(0) = 0$

$$P(1) = \frac{(1 + \sqrt{2})^1 - (1 - \sqrt{2})^1}{2\sqrt{2}} = 1$$

$$P(2) = \frac{(1 + \sqrt{2})^2 - (1 - \sqrt{2})^2}{2\sqrt{2}} = 2$$

Inductive step: For strong induction assume $P(n-1), P(n-2)$ is true for $n \geq 2$ and want to show that $P(n+1)$ is also true. By the definition of given sequence we know that,

$$\begin{aligned}
x_{n+1} &= 5x_{n-1} + 2x_{n-2} \\
&= 5 \left(\frac{(1+\sqrt{2})^{n-1} - (1-\sqrt{2})^{n-1}}{2\sqrt{2}} \right) + 2 \left(\frac{(1+\sqrt{2})^{n-2} - (1-\sqrt{2})^{n-2}}{2\sqrt{2}} \right) \\
&= \frac{1}{2\sqrt{2}} \left[\left(5(1+\sqrt{2})^{n-1} + 2(1+\sqrt{2})^{n-2} \right) - \left(5(1-\sqrt{2})^{n-1} + 2(1-\sqrt{2})^{n-2} \right) \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [7 + 5\sqrt{2}] - (1-\sqrt{2})^{n-2} [7 - 5\sqrt{2}] \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [1 + 3\sqrt{2} + 3(\sqrt{2})^2 + (\sqrt{2})^3] - (1-\sqrt{2})^{n-2} [1 - 3\sqrt{2} + 3(\sqrt{2})^2 - (\sqrt{2})^3] \right] \\
&= \frac{1}{2\sqrt{2}} \left[(1+\sqrt{2})^{n-2} [1 + \sqrt{2}]^3 - (1-\sqrt{2})^{n-2} [1 - \sqrt{2}]^3 \right] \\
&= \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}
\end{aligned}$$

Which implies $P(n+1)$ and by strong induction hypothesis $P(n)$ holds for all $n \geq 0$. □

Problem 3. Bounded Asymptotics

(a)

Proof. Function f is bounded from above therefore, $\exists B \in \mathbb{R} (B > 0)$ □